

SEDAR 11: Highly Migratory Species - Large Coastal Sharks

Data Workshop

Life History

- Whereas previous assessments have defined maximum sustainable yield (MSY) as 0.5 of carrying capacity, recent life history analysis and peer-reviewed literature has suggested this level is risk-prone, particularly for K-selected species (Musick et al. 2000). The life history group recommends a more conservative definition of MSY be adopted (i.e. 60-70% MSY or 40% of spawning stock biomass) for this assessment.
- Develop more empirical estimates of natural mortality for large coastal species.
- Research into further refining the separation of Gulf of Mexico stock of blacktip sharks using a combination of genetics, demography/life history and conventional and advance tagging technology (i.e. satellite archival tags).
- Updates on demographics using revised life history information.
- Continue research on life history characteristics of prohibited species.
- Research on stock-recruitment function for sharks.
- Accrue data necessary for ecosystem-based management: trophic relationships, bioenergetics, and diet.

Catch Statistics

- Biological data should be collected on the illegal Mexican shark catch confiscated in U.S. waters, including species, sex, and length.
- Gear-related information, including effort and gear used for each species should be collected on the interdicted Mexican vessels.
- One central electronic database for biological and gear data should be created to keep information regarding the confiscated sharks and vessels.
- Scientists should help the Coast Guard create the database and teach the agents how to identify the species and collect gear information.
- The Atlantic menhaden fishery data should be examined to determine shark bycatch estimates, if available.
- Historical data should be re-examined to determine if the “unreported catch” from Mr. Brannon is or is not already included in the commercial landings.
- Better landings information on number of species, by weight, from the dealers should be sought
- Dockside sampling information would be helpful to verify landings information such as species composition.
- Determine whether port sampler information for large coastal sharks is available and if so, how to access it.

Indices of Relative Abundance

- The working group recommended inverse weighting based upon CVs as the default weighting scheme whenever indices are not given equal weighting.

Assessment Workshop

The Assessment Workshop Panel identified the following Research Recommendations which will aid in future assessments.

- Data Workshop participants need to bring raw data to workshop to enable additional analysis to be conducted and reviewed during the workshop when practical
- Length frequency data should be provided when available, with particular reference to the VA LL dataset.
- Examination and analysis of the Pelagic Longline Observer data should be included.
- Identify nursery areas for sandbars in the northern Gulf of Mexico
- Additional life history studies for all complex species to allow for additional species-specific assessments.
- Additional life history research into sandbar sharks to supplement or replace the available data from the mid 1990s
- Incorporation of the University of North Carolina dataset collected by Dr. Frank Schwartz in the next LCS assessment, with recognition that it may also contain valuable information useful for the Small Coastal Shark assessment to be conducted in 2007.
- Examination of methods to incorporate tagging data information into the assessment
- Attempt to recover and quantify information on historic catch, with special emphasis prior to the 1993 FMP.
- Management to force contrast would improve the blacktip assessments.
- Additional length sampling and age composition collection to improve information for developing selectivities
- Initiation or expansion of dock side sampling for sharks
- Ensure that existing independent sampling programs be continued
- Ensure funding for the recently initiated (2002) pelagic survey being conducted by the Pascagoula laboratory- SEFSC

Review Workshop

Issue: Lack of species-specific data, and the inability to identify carcasses/logs/ fins to species level.

- Improve dockside monitoring of catches
- Increase observer coverage of the commercial fleet
- Use biochemical and/or genetic testing of products (carcasses/logs/fins) to produce reliable species identifications

Issue: Lack of life history data for some species within the large coastal shark species complex, which results in no meaningful estimate of intrinsic rate of increase (r) for use in assessments.

- Conduct research on the life history of all species in the complex, including regular sampling and analysis of the main species
- Use life tables (or other similar approaches) to estimate population parameters such as r

Issue: Limited numbers of longer term abundance (catch rate) data.

- Utilize all appropriate abundance series available, e.g. the Schwartz data from North Carolina

Issue: Geographic range of abundance surveys is variable, and those with limited geographic coverage are more likely to reflect localized changes than stock-wide changes.

- Evaluate alternative weighting schemes or modelling approaches for abundance data that take account of the geographic range of the surveys

Issue: Lack of species and size composition and effort data for abundance surveys.

- Provide information on species and size composition
- Obtain trends in deployed fishing effort at least for the catch-rate index series in Data Workshops and present them in the Assessment Workshop report, together with corresponding trends in catches and catch rate.

Issue: Information on the type and quality of the standardization used for abundance indices was not always available.

- Document the method of standardization used for all catch-rate indices
- Where possible, use the same standardization methods for all indices

Issue: Assessment of the Large Coastal Shark (LCS) complex does not represent the status of the stocks, or any particular component of the stocks.

- Develop species-specific assessments for the main components of the LCS complex, where possible. Continuing with the current approach will only result in confusion with regards to the status of these resources
- As an interim step, an improvement may be achieved if the complex can be split into smaller groups based on species with similar life history characteristics, or which occur within the same regions (e.g. the Gulf of Mexico or the Atlantic).